1 Components

The following sections describe the components (application plug-ins) that you can integrate in detail applications to display data.

1.1 Table view (grid)

The grid is used to display data in a tabular form. It provides several columns, which are grouped in so-called categories.

If you place a detail application of the "Table view" type on an application, all columns and categories are created automatically using the specified data source and according to the Data Description. On basis of this "default setting", you can then configure the grid.

Most configuration options are provided in the main context menu. Right-click a column header to open this context menu. If the Customizing mode is not enabled, some of the options are not available. The sections in the following mainly concentrate on the functions of this main context menu.

Grid properties

You can use the dialog "grid properties" to make advanced configurations for the column properties. To edit the properties of a column, select the column in the selection list (default is the column clicked on). The most important properties are:

- LanguageKey: Language key used for the column header. Important: Changes of the column header via the "Caption" property are invalid because they are overwritten with this configuration. Changes only become visible after closing and re-opening the application.
- Tooltip: Text displayed when the mouse is moved over the column header.
- FieldName: Name of the field from DataDescription, whose values are displayed in this column.
- SummaryItem: Formatting of column totals (see 0)
- BackColorField: Field name of column that specifies the color code for the background color. There are columns that include numerical color codes. If you specify their field names for a column, the respective color code is used for the background color of the column. You can use the configurator to hide the color column.
- ForeColorField: see BackColorField. Here, you specify the font color.
- HyperLinkField: Field name of column used for the hyperlink of this column. Some columns include URLs. If you enter their field name for a column, a double click on a field of the original column opens a URL. If the URL is a mail address, a "mailto" is automatically added. You can hide the actual hyperlink column during this process.

Moving columns/categories

You can move all columns and categories in the grid using drag and drop. Use the column configurator to hide columns or categories (main context menu, *Column chooser*). When the configurator is open, you can simply move the columns and categories that you want to hide. You can also use the configurator to show hidden columns and categories.

Sorting

To sort the grid by specific columns, simply click the header of the respective column. If you click the same column a second time, sorting is changed from ascending to descending. If you click a new column, the previous sorting is discarded unless the shift button is pressed at the same time.

You can also use the menu for sorting. The options *Sort ascending, Sort descending* and *Clear sorting* have the same effect.

Filter

In the grid, you can filter by specific values in the columns. Just click the pin next to the column header. A selection list of all values in this column is shown. This is the same procedure as filtering in MS Excel.

If you have set a filter for a column, this filter is displayed at the bottom left of the grid. Click the checkbox to remove the filter.

You can edit and refine the filter subsequently. For this purpose, the button "Edit filter" is shown at the bottom right of the grid. Click the button to open the filter editor that you can use to combine several filter criteria.

- Red text: Click the red text to open a list of possible linking methods for several filters.
- Blue text: Click to show a list of columns included in the grid. You can define a filter for these columns.
- Green text: Click to show the list of available comparators for a filter.
- Plus: A new filter is added.
- Cross: The respective filter is removed.

You can also open the filter editor via the menu.

Grouping

You can group the grid by one or several columns using drag and drop. To this end, drag a column header to the field on top of the grid

Right-click the "group by" field to open a dialog. *Maximize/Minimize* expands/collapses all groups. Click "Clear grouping" to delete the complete grouping. You can also drag and drop the column back to the grid.

You can also use the menu for grouping. Select *Group by this column*. You can change the visibility of the field by selecting *Show/hide group by box*.

Add/remove columns

You can manually add new columns to a grid. If you click *Add column*, a dialog opens. Enter the following values to create new columns:

- Name: Clear name of column
- Field name: Name of the field from DataDescription, whose values are displayed in this column.
- Header: Language key used for the column header.
- Category: Column category to which the column is to be allocated. The category must already exist. You can also move the column to another category subsequently.

Click *Remove column* to remove columns from the grid.

Add/remove categories

Use the menu function *Add category* to add new column categories. The name entered should be a valid language key in order to ensure correct translation.

Use *Remove category* to remove a category from the grid. Note: This is only possible with categories that do not include any columns.

Use *Rename category* to change the name of a category.

Fix categories

You can fix column categories on the left hand side of the grid using the menu function “Fix category”. This means that the fixed category is no longer affected when scrolling through the grid. This is useful if you want to permanently show one category, irrespective of the scrolling position.

The category might be moved to the left if it is fixed.

Column totals

In the grid, you can show the sum total for groups (*Show column totals for group*) and for overall columns (*Show overall column totals*).

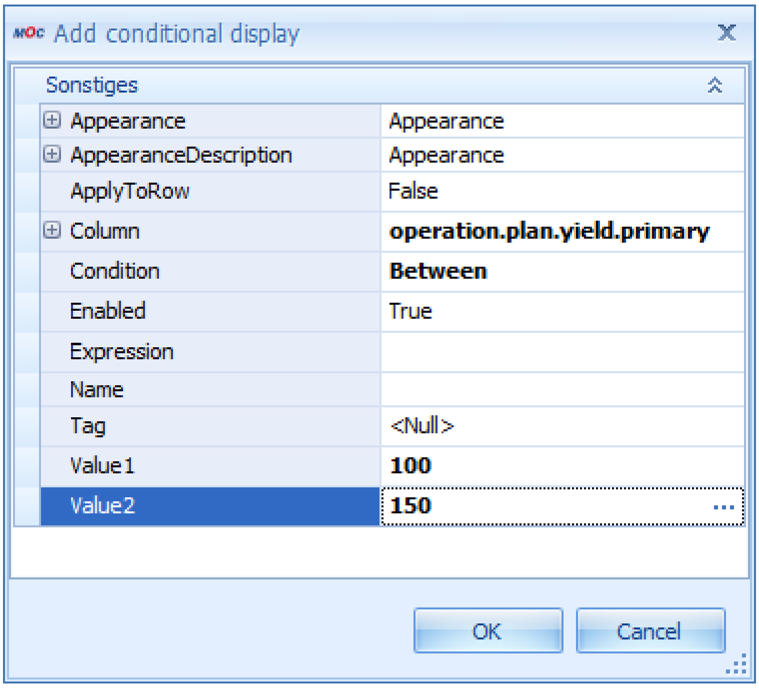
You can define the calculation of the sum totals via grid properties in *SummaryType* of the required column:

⊟ SummaryItem	**DevExpress.XtraGrid.GridColumnSummaryItem**
DisplayFormat	**{0:mpdv_timespan}**
FieldName	bookingrelation.operatorposition
SummaryType	**Custom**
Tag	**TimeSpanSum**

- DisplayFormat: Format string used to format the display. Possible formats are standard formats, e.g. {0:n0}, or the custom format {0:mpdv_timespan}. You can use the latter to specify a time for a value in seconds.
- FieldName: Name of the field from DataDescription, whose values are displayed in this column.
- SummaryType: there are different types of summaries. The most common ones are Sum, Count and Custom (see DisplayFormat and Tag). Tag: Only required if the custom format {0:mpdv_timespan} is used. In this case, TimeSpanSum must be entered here.

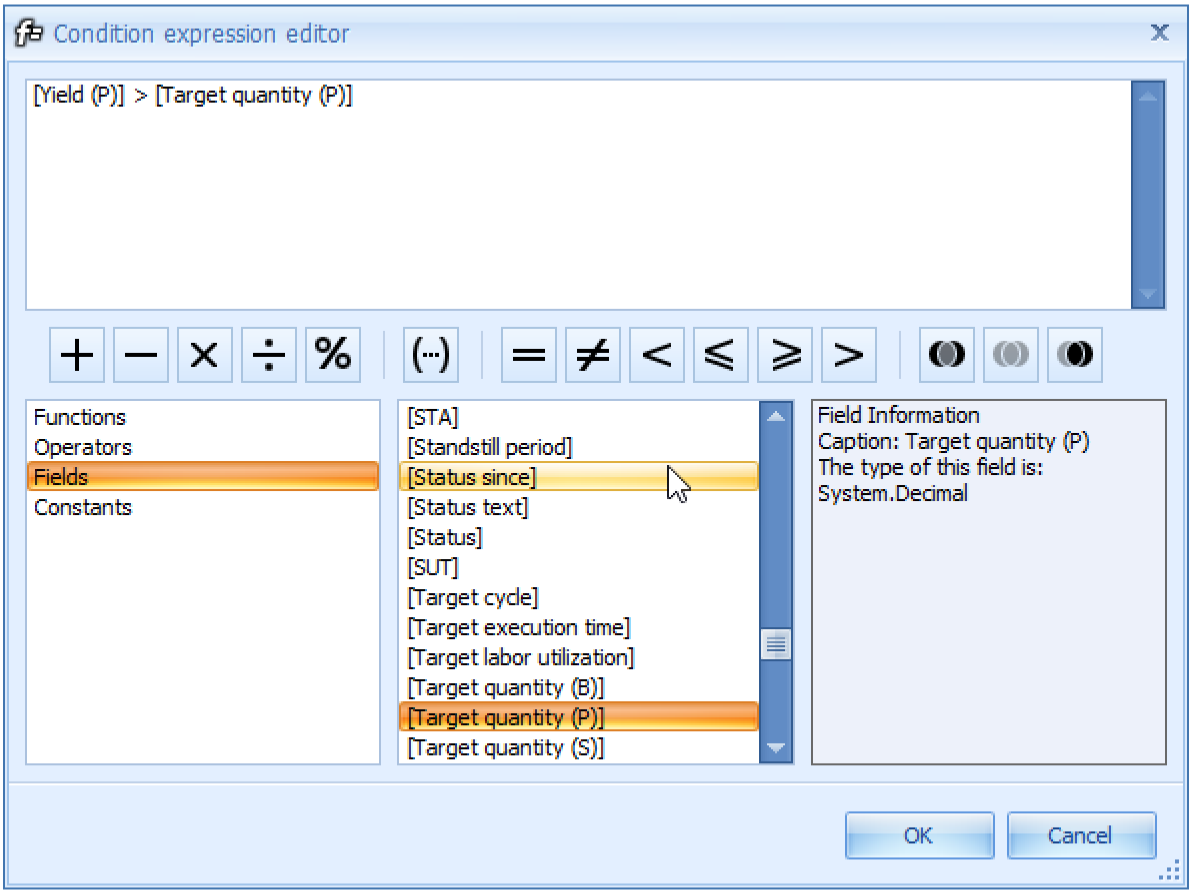
Conditional display

You can change the presentation of a grid column depending on the value in the cells. Open the dialog *Configure conditional display* that lists the conditions that are already configured. The name of a condition is composed of the major properties of that condition. Click *Add* or *Edit* to open the following dialog where you can edit a new or an existing condition:



In the image, all cells of the column operation.plan.yield.primary are colored in red, if their value is between 100 and 150.

- Appearance: Appearance summarizes all properties that affect the display of a grid column. For example, you can change the text type or the background color. If a cell of the selected column fulfills the condition defined below, the specified appearance is used for this cell.
- Column: Column the condition is defined for.
- Condition: Condition to be met.
- Value1: First comparative value.
- Value2: Second comparative value (e.g. if Condition = Between)
- Expression: You can define an expression that specifies the condition for the formatting. In this case, Expression must be specified for the Condition (Condition = Expression). You use the “Condition expression editor” to define an expression (to open the editor, click the three dots in field "Expression"):



The following objects are available for the definition of the expression:

- Functions: selection of mathematical functions and standard functions to edit date values, numeric values and character strings
- Operators: selection of logical operators
- Fields: available columns
- Constants: constant values

It is useful to create the required expression by double-clicking the different functions, operators, fields and constants, instead of editing them manually because this is less error-prone.

On the right hand side of the editor, the individual functions, operators, fields and constants are described.

An error message is displayed when the dialog is left if the expression defined does not meet the specified syntax.

Column width

You can change the column width if you simply "drag" the column. Move the mouse pointer over the edge of a column. The mouse pointer changes and the edge can be dragged to change the column width.

You can use the menu to set the optimum column width of a column (*Optimum column width*) or of all columns (*Optimum column width - all columns*). To identify the optimum width of a column, the system uses the contents of the relevant column.

Selection of Several Lines

If the menu item *Allow for several lines to be selected* is marked, the user can select several lines in the grid at the same time.

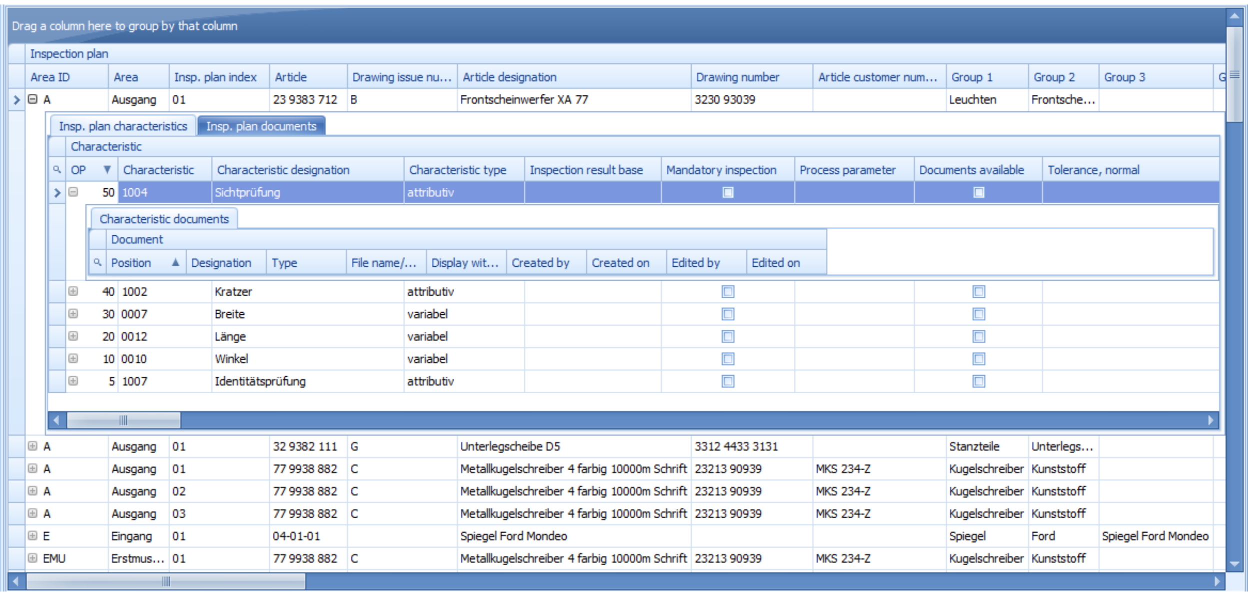
Configuration file

If a new application of the grid type is created, a separate configuration file is generated - as with most other detail application types, too. This file is stored in the application directory and is called Grid<Name_of_application>.config. All customizing possibilities described above have an effect on the contents of this file. Optionally, manual editing is also possible. The file includes the following sections:

- Layout: Layout of grid (columns, categories, grid-internal properties): This xml structure is automatically read by the DevExpress grid and must therefore have a specific structure.
- GroupPanelShown: specifies if the field ("group by") is displayed
- SummaryShown: specifies if the totals line is displayed
- GroupSummaryShown: specifies if totals lines for groups are displayed
- MultipleRowSelection: specifies if several lines can be selected at a time
- Styles: List of conditional formattings

1.2 Master Detail Grid

The Master Detail Grid is a hierarchical grid, i.e. the data is displayed on several hierarchical levels. The grid can include any number of levels. At each level (except the top level), there might be several grids.



The following example explains the structure of the Master Detail Grid and the steps that are required to configure a detail application for this grid.

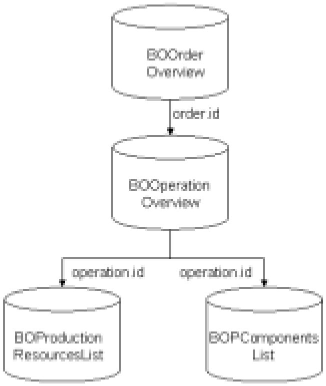
Basic data structure

The data source for a "normal" grid is a DataTable. Since a Master Detail Grid includes several levels, a DataSet, i.e. a collection of several interlinked DataTables is used here. The levels of a grid have a specific relationship to each other. Therefore, also the relations between the different DataTables within a DataSet must be specified.

The data for the grid are requested "on demand". This means that only the data for the top level are requested first. Clicking on the cross next to a data record will display another level. The data shown on this level is only requested when the level is expanded, indeed only the data displayed.

Example

The following sections follow this example: the first level shows orders, the second level shows operations. On a third level, production resources, tools and components are shown.



The image shows the structure of the DataSet with 4 DataTables from the example above.

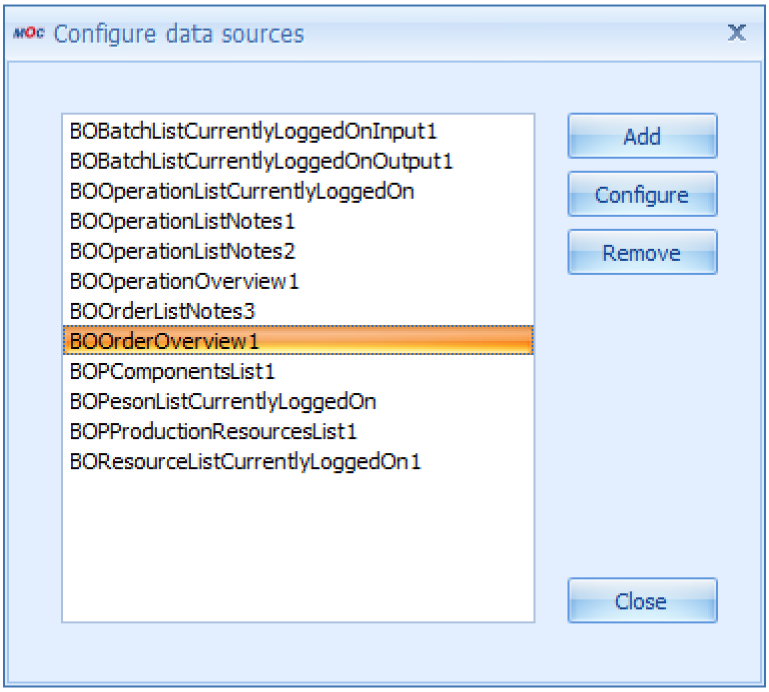
Clicking on the green arrow will consequently request all orders first. This data is inserted into the first DataTable (BOOrderOverview) within the Result DataSet. The order xyz is now expanded. All operations for this order are identified and added to the DataTable BOOrderOverview. If you also expand the order abc, the relevant operations are identified for this order, too, and the DataTable BOOperationOverview is added.

Inserting a detail application of type Master Detail Grid

If you want to add a new Master Detail Grid to an application, this application must be configured as follows:

Data sources

You must add data sources for each level of the grid.



The configured data sources of the example.

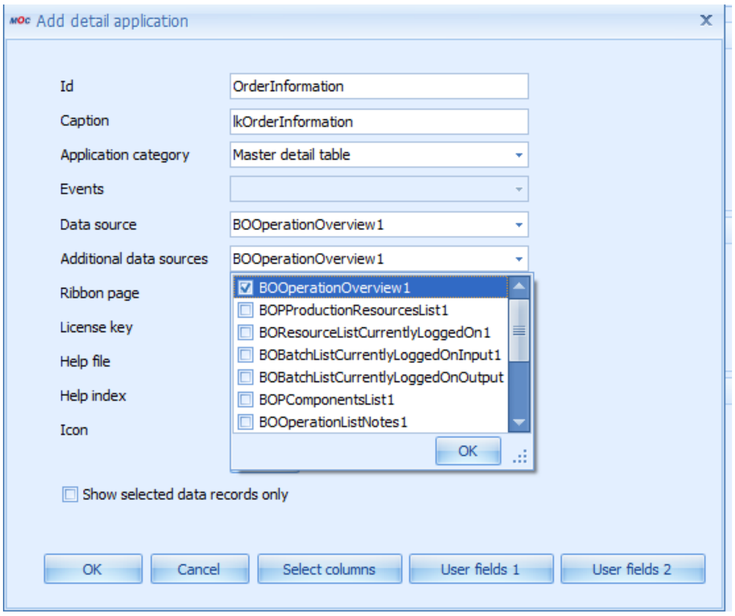
Note the following when you configure the individual data sources:

- The data source for the top level will usually react to a click on the "green arrow". The selection panel provides the required parameters. You must configure this accordingly.
- The inferior data sources must be configured as follows:
  - Events: Select the MasterRowExpanded event of the superior data source
  - Source: Select the superior data source
  - Parameter: Map all parameters that define the relationship between the two data sources (as in each master detail relationship)

Detail application

A new detail application must be added to the application. Configure as follows:

- Application category: Master Detail Table
- Data source: Select data source for top level
- Additional data sources: all other data sources whose data is to be shown in the grid



Configuration of the "Additional data sources" for the example.

Initializing the configuration file

If you create a new detail application (see step above) and you save the application, the system automatically creates a configuration file that includes the layout definition of this detail application. The name of this file is MasterDetailGrid<Name of detail application>.config.

The contents of a configuration file for a MasterDetailGrid are very similar to those of a "normal" grid. The only difference is that the settings are available several times – once for each detail grid. To uniquely identify the settings, the name of the respective data source is added.

See also: 1.1.

Important notes:

a) There are other, MDG-specific (MasterDetailGrid) properties that are missing in the configuration file of a "normal" grid. So far, side effects for the missing entry AllowExpandEmptyDetails are known. If this property is set to *false* (which unfortunately is the default setting), the MDG nodes are always only expanded upon the second click and only if data is available. The entry is as follows:

```
<property name="OptionsDetail" isnull="true" iskey="true">
   <property name="SmartDetailExpandButtonMode">AlwaysEnabled</property>
   <property name="AllowExpandEmptyDetails">true</property>
</property>
```

b) In order to hide the title line on lower levels (e.g. for reasons of space), the following must be added in the layout section of the superior level:

```
<property name="RowSeparatorHeight">0</property>
<property name="OptionsDetail" isnull="true" iskey="true">
   <property name="SmartDetailExpandButtonMode">AlwaysEnabled</property>
   <property name="AllowExpandEmptyDetails">true</property>
   <property name="ShowDetailTabs">false</property>
</property>
<property name="ColumnPanelRowHeight">-1</property>
```

The respective entry is ShowDetailTabs = false in OptionsDetail.

c) Tab label texts: by default, the label texts of tabs are made up of the translation of "lk" + name of the respective data source. Or: If you want a different label text, you can manually add an additional setting in the configuration file.

```
<Setting Key="TabTitle_xxx" Description="" LastChanged="2010-04-13T17:42:58.0115088Z" ValueType="System.String" Version="0.0.0.0">
  <Value>
    <string>lkTitle</string>
  </Value>
</Setting>
```

d) User fields: the user fields for each individual level must currently be added manually into the configuration file. Configure as follows:

```
<Setting Key="UserFieldObjectType_xxx" Description="" LastChanged="2011-07-25T18:49:05.2387406Z" ValueType="System.String" Version="0.0.0.0">
  <Value>
    <string>CPAN</string>
  </Value>
</Setting>
<Setting Key="UserFieldKey_xxx" Description="" LastChanged="2011-07-25T18:49:05.2387406Z" ValueType="System.String" Version="0.0.0.0">
  <Value>
    <string>FEP</string>
  </Value>
</Setting>
<Setting Key="UserFieldPraefix_xxx" Description="" LastChanged="2011-07-25T18:49:05.2387406Z" ValueType="System.String" Version="0.0.0.0">
  <Value>
    <string>inspectionrequirement</string>
  </Value>
</Setting>
```

Relations

In addition to the grid configuration for each level of the Master Detail Grid, you must define the relations between the levels. To this end, a further section is added at the end of the master detail configuration file. Here, there is NO editor and you must manually configure the file. For the manual configuration, proceed as follows (in the example):

```
<Setting Key="Relations" Description="" LastChanged="2009-07-07T11:52:21.229117Z" ValueType="System.Xml.XmlDocument" Version="0.0.0.0">
   <Value>
     <Relations>
```

```
        <Relation Name="Operations" MasterController="BOOrderOverview1"
DetailController="BOOperationOverview1">
          <ColumnRelations>
            <ColumnRelation>
              <ForeignKeyColumn DataSource="BOOrderOverview1"
Name="order.id"></ForeignKeyColumn>
              <KeyColumn DataSource="BOOperationOverview1"
Name="order.id"></KeyColumn>
            </ColumnRelation>
          </ColumnRelations>
        </Relation>
        <Relation Name="FHM" MasterController="BOOperationOverview1"
DetailController="BOProductionResourcesList1">
          <ColumnRelations>
            <ColumnRelation>
              <ForeignKeyColumn DataSource="BOOperationOverview1"
Name="operation.id"></ForeignKeyColumn>
              <KeyColumn DataSource="BOProductionResourcesList1"
Name="operation.id"></KeyColumn>
            </ColumnRelation>
          </ColumnRelations>
        </Relation>
        <Relation Name="Components" MasterController="BOOperationOverview1"
DetailController="BOPComponentsList1">
          <ColumnRelations>
            <ColumnRelation>
              <ForeignKeyColumn DataSource="BOOperationOverview1"
Name="operation.id"></ForeignKeyColumn>
              <KeyColumn DataSource="BOPComponentsList1"
Name="operation.id"></KeyColumn>
            </ColumnRelation>
          </ColumnRelations>
        </Relation>
      <Relations>
    </Value>
  </Setting>
```

Note:

- For each relation between two levels, you must insert one relation.
- You are free to select any name for the relation.
- MasterController is the name selected by the superior data source itself (the name must absolutely be identical).
- DetailController is the name selected by the inferior data source itself (the name must absolutely be identical).

- You must create a ColumnRelation for each relation between columns. ForeignKeyColumn is always the column in the superior data source, KeyColumn is the column in the inferior data source.
- As title of the inferior table in the MDG, "lk" + name of relation is always shown automatically. For this purpose, a language key usually has to be created in the mpdvDictionary. To hide this title (e.g. for reasons of space), change the following in the configuration file in the layout section of the relevant level: Set ShowDetailTabs in OptionsDetail to false (see example):

```
<property name="RowSeparatorHeight">0</property>
<property name="OptionsDetail" isnull="true" iskey="true">
   <property name="SmartDetailExpandButtonMode">AlwaysEnabled</property>
   <property name="AllowExpandEmptyDetails">true</property>
   <property name="ShowDetailTabs">false</property>
</property>
<property name="ColumnPanelRowHeight">-1</property>
```

Important: Make sure that ALL connections between columns are specified for each relation, which are required for an unambiguous assignment. Otherwise it can happen that too little data records are displayed in the grid (ambiguous data records cause an exception and this exception has the effect that the data record is not used for the display).

Before saving the (manually edited) file, you must shut down the MOC completely. Otherwise the relations, if any, are removed again!

Further configuration

The further configuration of the Master Detail Grid is similar to the grid configuration. But here, for each level the configuration is performed separately. All configuration options of the grid are available at each level of the Master Detail Grid. See also: 1.1.

Tips for troubleshooting

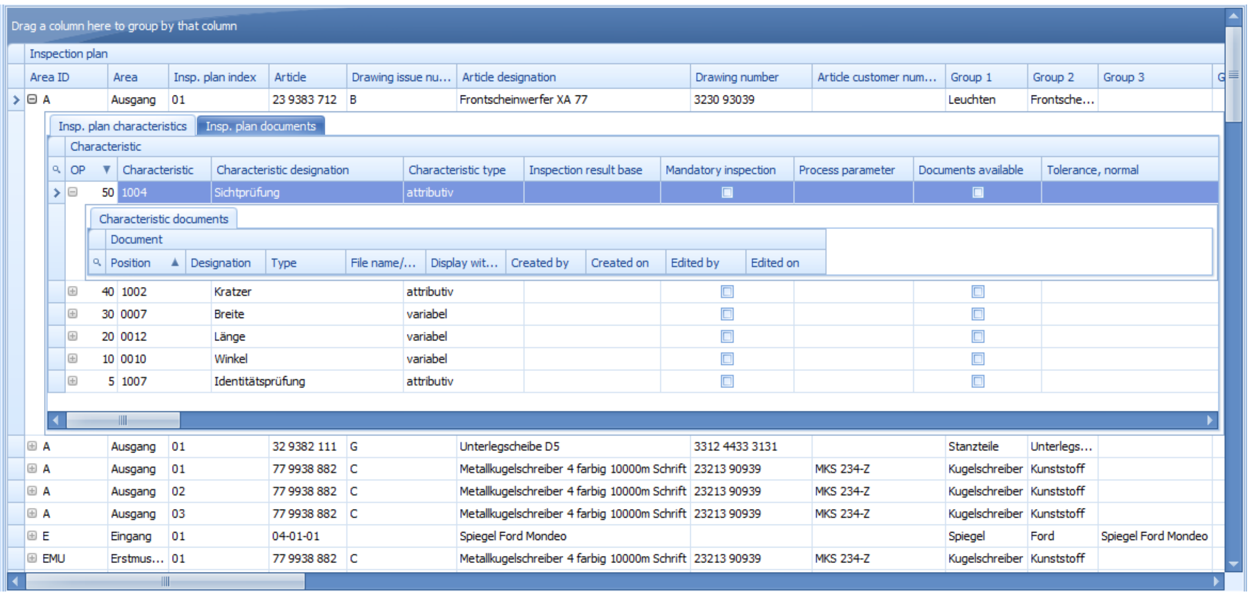
If data records are missing in one of the MDG levels, this can have two reasons:

- For the respective level, not all key columns have been defined that are required to uniquely identify a data record (in repository isKey = true).
- For the "Relations", not all connections between columns have been specified that are required to uniquely define the connection.

1.3 Master Detail Grid (without reloading)

The display and structure of the Master Detail Grid (without reloading) is similar to the "normal" Master Detail Grid.

It is a hierarchical grid, i.e. the data is displayed in several hierarchical levels. The grid can include any number of levels. At each level (except the top level), there might be several grids.



The section below explains the structure of this special form of Master Detail Grid (MDG) and the steps required to configure a detail application that is based on this MDG.

Basic data structure

The data source for a "normal" grid is a DataTable. Since a Master Detail Grid includes several levels, a DataSet, i.e. a collection of several DataTables is used here. The levels of a grid have a specific relationship to each other. Therefore, also the relations between the different DataTables within a DataSet must be specified.

The data is - and this is the difference to a "normal" MDG - NOT requested "on demand". The data is requested simultaneously for all grid levels e.g. when clicking the green arrow.

Reason for this behavior: For this detail application of the Master Detail Grid type (without reloading), a data source has been defined or selected, that provides a DataSet as result. Important: This must be discussed with the system designer in advance.

<u>Special case:</u> in exceptional cases it is possible, e.g. by script, to compose such a DataSet manually from the results of several data sources. An example for this is the PaymentDayTypes application with its related scripts (see also 0).

In the repository, the structure of the result DataSet of a data source is defined on the ServiceParameters tab. The ResultSet column is used to note the level where each column is located. In the result DataSet, there will later be a DataTable for all entries under ResultSet in the repository that can be used for one level of the MDG.

Inserting a detail application of type Master Detail Grid (without reloading)

If you want to add a new Master Detail Grid to an application, this application must be configured as follows:

Data sources

You must define 1 data source (providing a DataSet as result).

Special case: If such a data source is not available but if data from several different data sources (whose data source is DataTable) are presented in the grid, these data sources must be combined into a DataSet via script and assigned to the application (see also 0).

Detail application

You muast add a new detail application of type "Master detail table (without reloading)" to the application. Configure as follows:

- Application type: Master Detail Table (without reloading)
- Data source: Select suitable data source (in case of more than one data source, specify the data source for the top level)

Creating/Initializing the configuration file

You can use a configuration file to define the layout of this detail application. For technical reasons, this is NOT created automatically for the MDG (without reloading). How such a file may be created and edited is explained below.

First create a text file with the name MasterDetailGridWithoutReload[name of detail application].config. Save this file in the application directory of the main application.

The configuration of a normal grid with only one level is made up of several sections (see 0). The configuration of an MDG is made up of the configuration of such a normal grid for each level. Consequently, each level must be configured separately[1].

You need not edit these configurations manually. Proceed as follows:

- Create an empty test application.
- Add the data source to the test application that is also used for the MDG. Important: Select all columns in the column configurator that are required in the relevant MDG level.

---

[1] If several grids are located in one level, one configuration each must be available for the different grids. In this case, several configurations for one level might exist.

- Add a normal grid to the test application as detail application. Configure the grid, if required.
- The last two steps are repeated for all levels of the MDG. Important: The selected columns of the data source must always have the same entry under ResultSet in the repository.
- Now save the application. For each grid, a configuration file with the name Grid<Name of detail application>.config is created.
- Copy the contents (= all sections) of all configuration files into the previously created MDG configuration file (one below the other).
- Rename all sections as follows:
  <Name of section>_<Associated ResultSet from repository>

Important notes:

a) The layout section in the configuration file of a "normal" grid usually DOES NOT contain any entry for the property Name. This, however, results in errors when requesting data within an MDG. The name must absolutely correspond to the name of the associated ResultSet (see Repository). For this reason, manual modifications are required if such a layout section is copied to the configuration of an MDG.

b) There are other, MDG-specific properties also missing in the configuration file of a "normal" grid. So far, side effects for the missing entry AllowExpandEmptyDetails are known. If this property is set to *false* (which unfortunately is the default setting), the MDG nodes are always only expanded upon the second click and only if data is available. The entry is as follows:

```
<property name="OptionsDetail" isnull="true" iskey="true">
   <property name="SmartDetailExpandButtonMode">AlwaysEnabled</property>
   <property name="AllowExpandEmptyDetails">true</property>
</property>
```

c) In order to hide the title line in lower levels (e.g. for reasons of space), the following must be added in the layout section of the superior level:

```
<property name="RowSeparatorHeight">0</property>
<property name="OptionsDetail" isnull="true" iskey="true">
   <property name="SmartDetailExpandButtonMode">AlwaysEnabled</property>
   <property name="AllowExpandEmptyDetails">true</property>
   <property name="ShowDetailTabs">false</property>
</property>
<property name="ColumnPanelRowHeight">-1</property>
```

The respective entry is ShowDetailTabs = false in OptionsDetail.

Relations

In addition to the grid configuration for each level of the Master Detail Grid, you must define the relations between the levels. To this end, a further section is added at the end of the master detail configuration file. Here, there is NO editor and you must manually configure the file. For the manual configuration, proceed as follows (in the example):

```
<Setting Key="Relations" Description="" LastChanged="2009-07-07T11:52:21.229117Z"
ValueType="System.Xml.XmlDocument" Version="0.0.0.0">
  <Value>
    <Relations>
       <Relation Name="order_operation" MasterController="order" DetailController="operation">
          <ColumnRelations>
            <ColumnRelation>
              <ForeignKeyColumn DataSource="order" Name="order.id"></ForeignKeyColumn>
              <KeyColumn DataSource="operation" Name="order.id"></KeyColumn>
            </ColumnRelation>
          </ColumnRelations>
        </Relation>
      </Relations>
    </Value>
  </Setting>
```

Note:

- For each relation between two levels, you must insert one relation.
- You are free to select any name for the relation.
- MasterController is the name of the superior ResultSet (the name must absolutely be identical).
- DetailController is the name of the inferior ResultSet (the name must absolutely be identical).
- You must create a ColumnRelation for each relation between columns. ForeignKeyColumn is always the column in the superior data source, KeyColumn is the column in the inferior data source.
- Before saving the file, you must shut down the MOC completely. Otherwise the relations, if any, are removed again.

Tables

You must also define all levels. To this end, a further section is added at the end of the master detail configuration file. Here, there is NO editor and you must manually configure the file. For the manual configuration, proceed as follows (in the example):

```
<Setting Key="Tables" Description="" LastChanged="2009-
  <Value>
    <Tables>
      <Table>order</Table>
      <Table>operation</Table>
    </Tables>
  </Value>
</Setting>
```

Note: For each level/each grid, you must add an entry. The specified name matches the name of the associated ResultSet.

MasterTable

In addition, the name of the main level must be defined. To this end, you attach a further section at the end of the Master Detail configuration file. Here, there is NO editor and you must manually configure the file. For the manual configuration, proceed as follows (in the example):

```
<Setting Key="MasterTable" Desc
  <Value>
    <string>order</string>
  </Value>
</Setting>
```

Note: The name specified must match the associated ResultSet.

Further configuration

You should have made the configuration of the individual grid levels directly for the underlying grids, because for technical reasons it is not possible to save them. Subsequent changes must directly be made in the configuration file.

Special case: Composing DataSet from results of several data sources

There are 2 possibilities if no data source with several results (as DataSet) exists, but if the data is nevertheless to be presented in an MDG:

- Either you use a regular MDG with reload of data on demand (description in 1.2);
- Or you use the MDG without reload of data and compose the results (as DataTable) manually via script.

Note the following for the second variant:

- Currently, the only possibility to create the new ResultSet is via script. The respective script entry point is <Name of detail application> + `AltSetDataSource`.
- You must specify the inferior data sources as additional data sources for the detail application (as for the normal MDG).
- The name of the sections in the configuration file is the name of the data source (as for the normal MDG).

An example for this procedure is included in the configuration file of the MDG and/or in the script for the PaymentDayTypes application.

1.4 Charts

The ChartControl (Chart) is a control tool to visualize data. The chart may handle different types of data sources. In addition, a wizard is provided allowing for all major settings being made by the user and/or application designer.

Available chart types

Chart: Standard chart, completely configured through ChartWizard.

Chart with series selection: See chart, but the user can show and/or hide individual series.

Individual series chart: Special chart to display data of several columns of one data source in a pie chart. All columns configured in the column configurator are consecutively written into a series. The column header is used as an argument (label). The configuration is made using the ChartWizard.

RPA distribution: See individual series chart. RPA colors are provided as palette. Using a radio button, the user can decide if the tooltip displays times or quantities.

Cycle progression: Special chart to display a cycle progression. Data source and series are pre-configured and cannot be edited.

Downtime hit list: Special chart to display a downtime hit list. Data source and series are pre-configured and cannot be edited.

Article profile: Special chart to display an article profile.

PDV measurement analysis: Special chart to display the PDV measurement analysis.

ChartWizard

You can use the ChartWizard to edit most settings of the chart application graphically. Note: Only if the application has already requested data, a selection of data series in the wizard is possible.

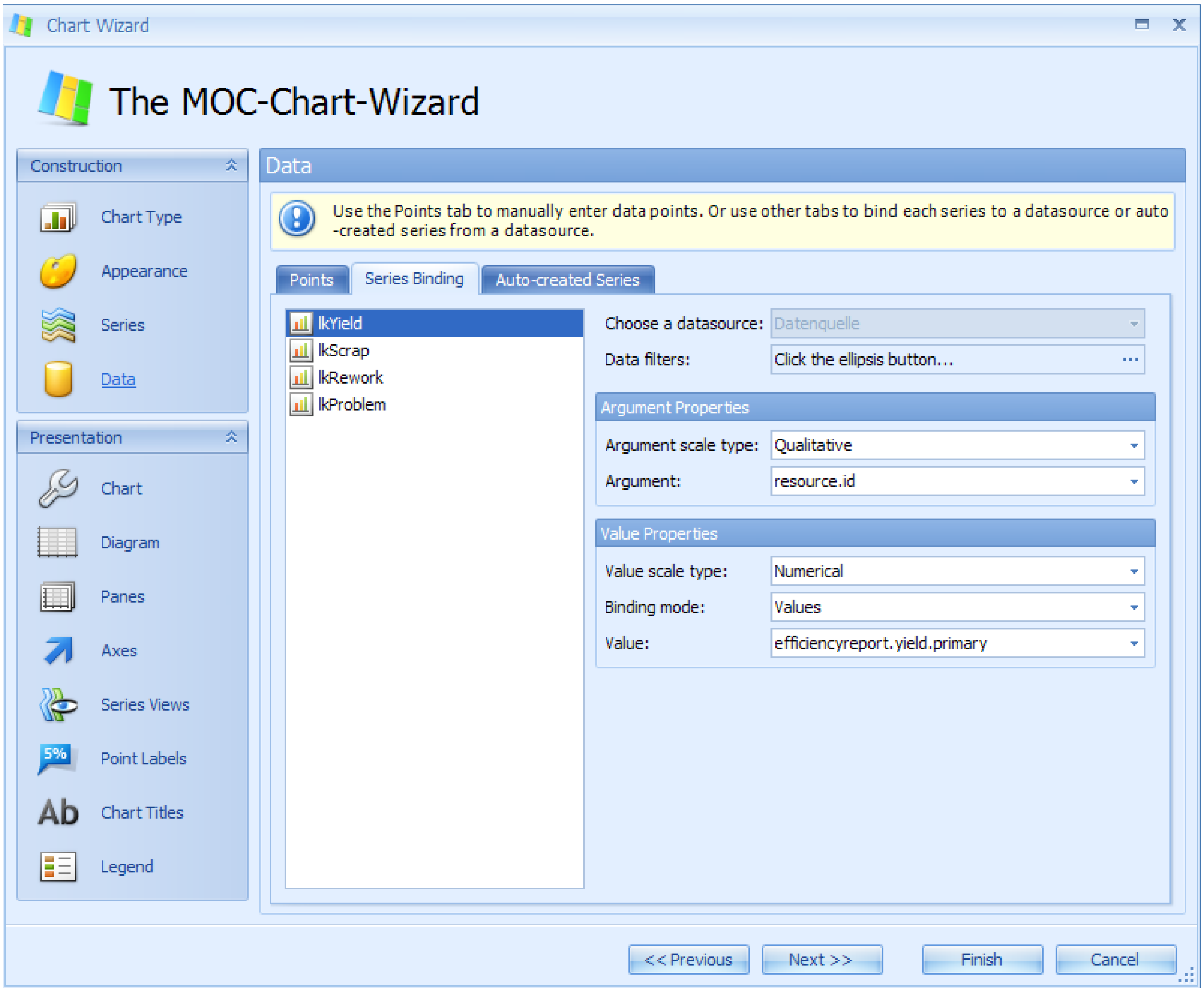
To activate the wizard, double-click a chart. The following setting options are available:

Chart type: Selection of chart type, e.g. bar chart, pie chart

Appearance: A color scheme may be selected and a color palette may be specified.

Series: Creating series. The data types of the axes must be specified. The series names must be language keys.

Data: Binding series to the fields of the linked data source.



The figure shows a sample configuration in the ChartWizard.

1.5 Layout

A detail application with the category "Layout" is used to display the contents of individual data records in fields (controls), which may be distributed among several tabs of the application. The configuration of the application is similar to the one of the selection panel.

Detailed information on maintenance is included in the sections "Configuration of applications - Selection panel" and "Input and output fields".

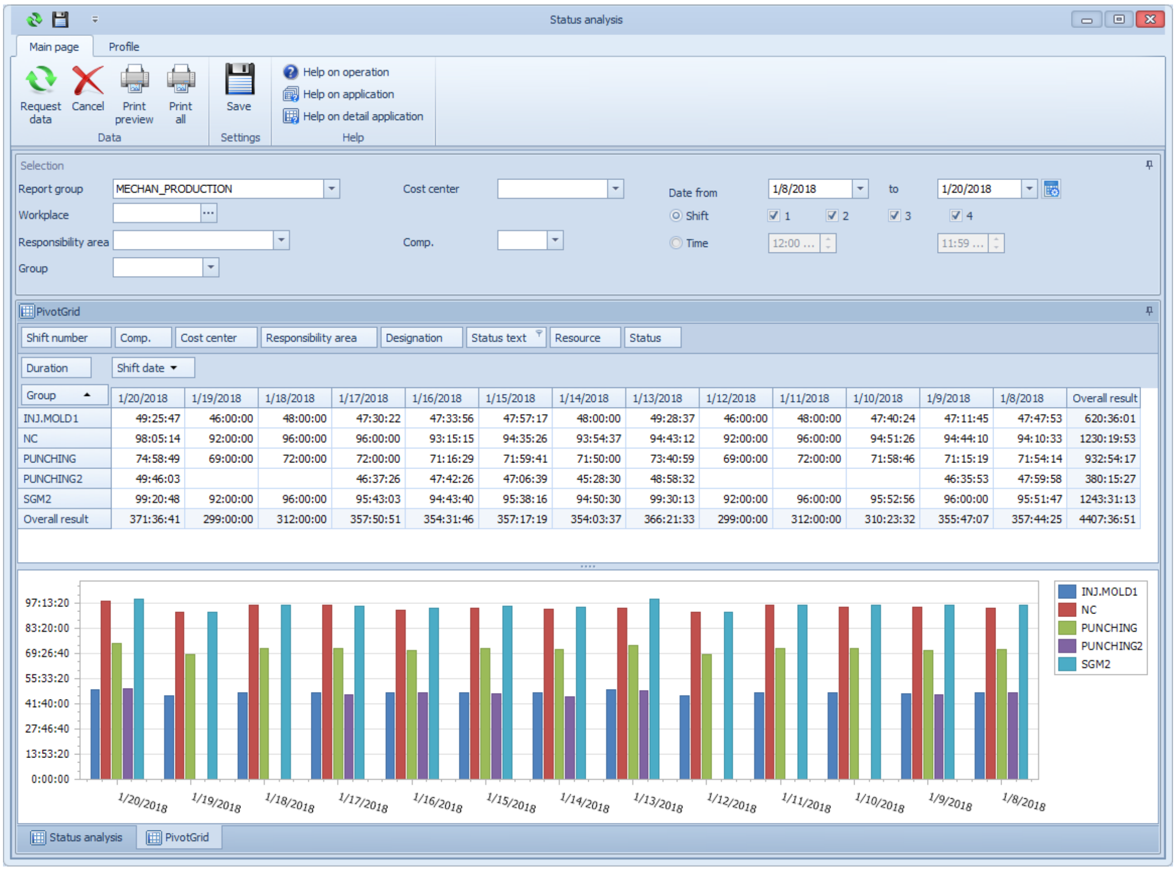
Special notes

- Select "Show selected data records only" in the detail application configuration.
- The values transferred are based on the data source. The key here is the FieldName (provided by WebService, data source and must comply with the FieldName of the InputControl).
- Set the property "ShownInDetail" = "true" for fields that you want to display in application details.
- Use an entry in "Category 1" to show fields in a specific tab.

Many changes in InputControls will only be visible upon saving, closing and re-opening the application!

1.6 Pivot

Using the pivot module, you can show summaries of data from a simple table in a pivot table using the relevant columns. You can use different aggregate functions (sum total, mean value, etc.) to calculate the contents of the created table. You can place the available columns in the four areas at runtime using drag and drop in order to highlight different focuses or to identify trends.



Areas of the pivot table

The header area (where the filter fields are placed) is divided into:

- **Row Header Area**, field "Group" in the example
- **Column Header Area**, field "Shift date" in the example
- **Data Header Area**, field "Duration" in the example. Calculations are made after this field.
- **Filter area**, fields "Shift number", "Company", etc. in the example. The fields in this area may be used in order to make further restrictions, e.g. calculation of durations for a particular status only (to be selected via "Status text" field).

In the actual **Data Area**, the aggregate data is presented, including the total results according to rows and columns, as well as intermediate results if several fields are placed in the row or column area. In the example, the duration is summed up according to group and shift date.

Field list

Using the column selection of the data source of the detail application, the application designer specifies the fields that are available in the pivot detail application (according to the columns of a flat table). The field list provides these fields for the user. From there, you can transfer and/or remove the fields into or from the four header areas. The remaining fields are not involved in the calculations, but are available to the user for further compositions.

Context menus

The following different context menus are provided for the pivot table:

Context menu in empty header field (main menu)

- Show/hide all fields: all fields in the field list are added to the pivot table as fields and/or all fields of the pivot table are removed from all areas of the table.
- Hide filter area fields: all fields are removed from the pivot table filter area.
- Hide/show field list: hide/show selection list with available fields.
- Show FilterEditor: the FilterEditor may be used to implement additional filters for the pivot table.
- Show/hide settings: this menu item may be used to open an additional editing dialog (please also refer to "Settings" below).

Settings

- Top N: specifies how many data is shown in the table. Note: this setting only works if "Top N" is activated for data fields (see below) in the context menu.
- Show others: if the option "Top N" is activated, you can use this option to specify if the other data is shown in the table as summarized form.
- Totals location: specifies where the totals columns are shown in the table. Far: right; Near: left
- Chart: Is the chart area of the pivot shown?
- Selection: Is only selected data shown in the chart?
- Columns: Are the values in the chart shown in different columns or in "cumulative" form?
- Totals: Are totals additionally shown in the chart?
- Legend: Is the legend shown in the chart?

Context menu in row and column fields

- Show/hide subtotals: is used to specify if only totals or also subtotals for individual groups are shown in the table.
- Hide: This is used to remove the selected field from the relevant area.
- Sequence: This is used to specify the sequence of fields in the relevant area.
- Optimum column width / Optimum column width (all columns): see also 0. Important: in the pivot table, all columns always have the same width.

Context menu in data fields.

- Aggregate functions: is used to specify how data is aggregated in the pivot table. The following functions are available: Total, Number, Minimum, Maximum, Variance, Variance (Percent), Standard deviation, Standard deviation (Percent). Important: It is possible that specific aggregate functions are not useful depending on the data formatting.
- Additional aggregate function: The Ratio aggregate function is based on the single values of two adjacent columns (dividend and divisor).
- Example from Time and Attendance PZE: Actual time=99, target time=100. Ratio=0,99, displayed (via OutputFormat) as %actualtime/targettime=99%.
- What is special about this function is that the column values are not based on their own single values and cannot be computed from other, aggregate values.
- What is also special about the calculation: in case of a division by 0, the result is set to 0.
- Show aggregate function name: if this option is selected, the name of the selected aggregate function is displayed behind the data field.
- Display type: specifies how the data is displayed. Options:
    - Nominal: the value is displayed (default)
    - Difference (absolute): the difference to the previous value is displayed.
    - Difference (%): the percentage difference to the previous value is displayed.
- Hide: Removes field from data list
- Sequence: see above
- Optimum column width / Optimum column width (all columns): see also 0.
- Sort ascending/descending: specifies sorting of data
- Top N: specifies whether only the top n data records are shown. In addition, the settings dialog may be used to specify n.

Associated chart

The associated chart graphically presents the current data of the pivot table. An arbitrary row/column selection may also be displayed. See also "Pivot table settings". In the Customizing mode, you can configure the chart in more detail using the ChartWizard (double-click to open).

Pivot table settings

The panel for the table settings and the associated chart may be shown and hidden via context menu in the empty header field.

The panel with settings will be migrated to the application menu in a future version.

1.7 FileAttachmentPlugin

Overview

You can use the FileAttatchmentPlugin to assign a file as attachment to an existing data record in a grid. The FileAttachmentPlugin can be used for the links in the toolbar. The file is uploaded to the server into a configurable path. You can also use the plug-in to show the file, to delete the file and to remove it from the data record.

Requirements

1) You must configure a path in the path configuration where the files are stored. You can also use an already configured path.
2) The data records in the grid must have a column with a unique key. This key then becomes part of the file name on the server. Example: internal ID (serial).
3) The data records in the grid must include a column of data type string. The name of the file that is uploaded to the server is then written into this column. If no attachment is assigned, this column must be empty. When you assign an attachment, the plug-in automatically populates this column using a generated file name. If you delete an attachment, the plug-in removes the file name from the data record.

Assigning an attachment

Configuration in the link editor

Command

Fixed "UploadAttachment"

Function

Fixed "callCommandObject"

Parameter

Example:

UploadAttachment hydraPath="MOCHRIMG"
attattchmentFileName="pequal_[personnelqualificationsassignment.internal_id]"
listDataLogic="PersonnelQualificationsAssignmentList"
updateDataLogicId="PersonnelQualificationsAssignmentUpdate"
updateDataLogicParameter="personnelqualificationsassignment.person.id,personnelqualification
sassignment.qualification.id,personnelqualificationsassignment.internal_id,personnelqualifications
assignment.valid_from,personnelqualificationsassignment.valid_to"
updateDataLogicField="personnelqualificationsassignment.filename"

Parameter	Explanation

UploadAttachment	Fixed CommandLinkID
hydraPath	HYDRA path used to upload the files.
attachmentFileName	Column that includes the name pattern for the file name on the server, e.g. "pequal_[personnelqualificationsassignment.internal_id]"
listDataLogic	Data source of list service
updateDataLogicId	Data source of update service
updateDataLogicParameter	List of columns that must be passed to the update service as parameters.
updateDataLogicField	Column of update service that includes the file name.

Other fields

For the other fields, e.g. Language Keys and symbols, nothing special applies.

Process description

1. If you click the icon to upload an attachment in the toolbar, a dialog opens where you can select a file.
2. The plug-in generates a unique file name for the file on the server according to the configured pattern (parameter attatchmentFileName). The file extension of the original file is not changed.
3. The plug-in then copies the file with the generated name into the configured path.
4. As a next step, the selected data record is automatically locked by the lock service (key like update service).
5. The plug-in calls the update service to save the generated file name in the configured column of the data object. The plug-in passes the configured parameters from the list service as key to the update service and also the configured column with the file name.
6. The plug-in then automatically performs the unlock service (key as update service).

Show attachment

Configuration in the link editor

Command

Fixed "ShowAttachment"

Function

Fixed "callCommandObject"

Parameter

Example:

ShowAttatchment hydraPath="MOCHRIMG"
listDataLogic="PersonnelQualificationsAssignmentList"
updateDataLogicField="personnelqualificationsassignment.filename"

Parameter	Explanation
ShowAttachment	Fixed CommandLinkID
hydraPath	HYDRA path used to upload the files.
listDataLogic	Data source of list service
updateDataLogicField	Column of service that includes the file name.

Other fields

For the other fields, e.g. Language Keys and symbols, nothing special applies.

Process description

1. If you click the icon to show an attachment in the toolbar, the file is automatically downloaded to the client. (The MOC stores the files in the directory of application data of the user; the files are automatically deleted when the user logs off or when the MOC is shut down).
2. The plug-in opens the file using the operating system. To show the file, the application is called that is assigned to the respective file extension. If no application is assigned to the file extension, Windows provides a selection of possible applications.

Delete attachment

Configuration in the link editor

Command

Fixed "DeleteAttatchment"

Function

Fixed "callCommandObject"

Parameter

Example:

DeleteAttatchment hydraPath="MOCHRIMG"
listDataLogic="PersonnelQualificationsAssignmentList"
updateDataLogicId="PersonnelQualificationsAssignmentUpdate"
updateDataLogicParameter="personnelqualificationsassignment.person.id,personnelqualificationsassignment.qualification.id,personnelqualificationsassignment.internal_id,personnelqualificationsassignment.valid_from,personnelqualificationsassignment.valid_to"
updateDataLogicField="personnelqualificationsassignment.filename"

Parameter	Explanation
DeleteAttachment	Fixed CommandLinkID

hydraPath	HYDRA path used to upload the files.
listDataLogic	Data source of list service
updateDataLogicId	Data source of update service
updateDataLogicParameter	List of columns that must be passed to the update service as parameters.
updateDataLogicField	Column of update service that includes the file name.

Other fields

For the other fields, e.g. Language Keys and symbols, nothing special applies.

Process description

1. If you click the icon to delete an attachment in the toolbar, a confirmation prompt opens.
2. If you click OK to confirm, the plug-in deletes the file with the name from the configured column of the list service in the configured path.
3. As a next step, the selected data record is automatically locked by the lock service (key like update service).
4. The plug-in calls the update service to set to empty the file name in the configured column of the data object. The plug-in passes the configured parameters from the list service and the configured column with the empty file name to the update service.
5. The plug-in then automatically performs the unlock service (key as update service).

1.8 Special table views

The following sections describe other table views used in very special application cases.

The special table views are intended for internal use by MPDV itself. The special table views have been developed for very specific use cases and it is normally not useful to use them in other situations. They require extensive knowledge of the configuration files on the MOC. A full use might only be possible if an additional programming (scripting) is performed. This is not possible with the tools that are available for customers.

Year model

The year model is a special grid to display and edit data in a calendar form. This type of detail application was especially developed for the year model application and is used to maintain data. Apart from the grid itself, it contains additional controls and input fields. Of these, year, model and designation/name as well as the grid contents are actually used for data maintenance. The fields below the grid are only used for editing the grid at runtime.

The year model is used for data maintenance and is therefore only placed on maintenance applications. As the data source, MDMFYearModelInsert or MDMFYearModelUpdate may be used optionally. The column configurator is empty. Further configuration is not required.

PDV ID Tracing

The grid for PDV ID tracing is a grid especially used for the ID tracing application. The only difference to the "normal" grid is that the columns are generated dynamically at runtime. The columns stored in the configuration are NOT integrated.

The columns `resource.id` and `pdvtagbasedsinglevalue.capture_ts` are permanently added at runtime. Depending on the data provided by the web service, more columns are added.

For this plug-in, only the data source `PdvTagBasedSingleValueList` can be used. In the column configurator, all columns should be selected.

The configuration is identical to the one of the grid, see 1.1.

PDV process analysis

The grid for a PDV process analysis is a grid especially used for the process analysis applications (machine-related, order-related, batch-related). The only difference to a "normal" grid is that the measured value that is displayed is taken from one of 4 different columns. The data type specifies which of the four columns is used.

For this plug-in, only the data source `PdvSingleValueListWithAdditionalInfo` can be used. In the column configurator, all columns should be selected.

The configuration is identical to the one of the grid, see 1.1.

1.9 Special charts

The following sections describe other chart types used in very special use cases.

The special charts are intended for internal use by MPDV. The special charts have been developed for very specific use cases and it is normally not useful to use them in other situations. They require extensive knowledge of the configuration files on the MOC. A full use might only be possible if an additional programming (scripting) is performed. This is not possible with the tools that are available for customers.

PDV measurement analysis

The chart for the PDV measurement analysis is a special chart that can only be used for special applications: Applications for the measurement analysis of measured values and samples, as well as for machine-related, order-related and batch-related measurement analyses.

This plug-in can include up to 3 charts that are added/removed using . For each chart, several process parameters can be selected from the associated selection list. Machine/order/batch and time range specify the selection available in the list. Initially, the list is empty.

When configuring an application using this plug-in, note the following:

- For single values, limit values and events, you must define 3 data sources each; this means that 9 data sources must be defined in total.
- Permitted data sources are:
    - Single values: PdvSingleValueList, PdvSampleValueList
    - Limit values: PdvSingleValueSpecification, PdvSampleValueSpecification
    - Events: PdvEventList
- For each data source, all columns in the column configurator should be selected.
- For the detail application, all data sources must be set. The first data source of single values is the main data source, the remaining ones are additional data sources.

1.10 Grouping application

The grouping application is intended for internal use by MPDV. It requires extensive knowledge of the configuration files on the MOC. A full use might only be possible if an additional programming (scripting) is performed. This is not possible with the tools that are available for customers.

The grouping application does not have any data source but - as is suggested by its name - is used to group other detail applications. For this reason, it is not necessary to create a data source before a grouping application is placed on an application. After selecting the application type, an appropriate information window is therefore shown.

To add other applications to the group, double-click the application. This is only possible if the MESDevelopmentSuite is activated. The grouped applications must be docked within the application. A dialog opens. The left hand side shows all existing detail applications of the main application. The right hand side shows the list of all detail applications already included in this grouping. Applications can be moved by clicking the arrow buttons.

If detail applications are moved to the group, you can dock them again within the group. After docking, the application must be restarted. You can also use this procedure in the inverse direction, to move applications back from the group to the main application.